9.3. GAS COUNTERS

All gas-filled counters are in principle ion chambers (with the exception of the less common gas scintillation counters). The ionization produced in an ion chamber by a single nuclear particle produces too low a charge pulse to be easily detectable except for α- particles. However, an ion chamber can be designed so that the number of ion pairs formed in each event is multiplied greatly.

Consider an ion chamber with a hollow cylindrical cathode and a thin central wire as an anode (Fig. 9.9(a) shows an old type GM-tube). The ion pairs formed in the gas by the passage of the ionizing radiation are separated from each other by their attraction to the electrodes. The small, very mobile electrons are rapidly collected on the anode, which is maintained at a high positive potential above ground, $\geq$1000 V. Most of the voltage pulse which appears on the anode arises by induction by the positive ions as they move away from the immediate region of the anode. This step is responsible for the rise time of the pulse, curve I in Figure 9.3(b). The potential decrease is only momentary as the anode is rapidly recharged by the power supply. The time necessary to restore the original potential is a function of the decrease in electric field near the anode due to the build-up of a layer of slow-moving positive ions, and of the time constant circuitry (curve II in Fig. 9.3(b)).

The electric field strength at a distance x from the anode is proportional to $1/x$. If the applied voltage of a cylindrical chamber of 1 cm radius is 1000 V, the potential in the immediate vicinity of a center wire of 0.0025 cm diameter is approximately 7×10^4 V cm^{-1}. As the primary electrons reach the vicinity of this high field and increase their kinetic energy, they cause secondary ionization which increases the pulse detected at the wire anode. The collected charge is given by Eqn (9.7), where a, *the gas multiplication factor*, is $\gg 1$.

This gas multiplication factor varies with the applied voltage as illustrated in Figure 9.8. In region II of the curve a flat plateau over a relatively wide voltage range is observed. Prior to the attainment of the threshold voltage for the plateau, the ions would not have sufficient *drift velocity* to prevent elimination of some ion pairs by recombination. Throughout the plateau range the drift velocity is sufficient to make recombination negligible and since secondary ionization is not present, $a = 1$. Ionization chambers operate in this voltage region. In region III the electrons from the primary ion pairs receive sufficient acceleration to produce additional ionization and the process of gas multiplication ($a > 1$) increases the number of collected charges. This is the region of *proportional counter operation* as the pulses are proportional in size to the energy deposited in the detector by the passage of the initial radiation. Region IV is the one used for *Geiger-Müller counter* (or GM counter) operation. In this region the gas multiplication is very high ($\geq 10^6$) and the pulse size is completely independent of the initial ionization. Beyond region IV continuous discharge in the detector occurs.

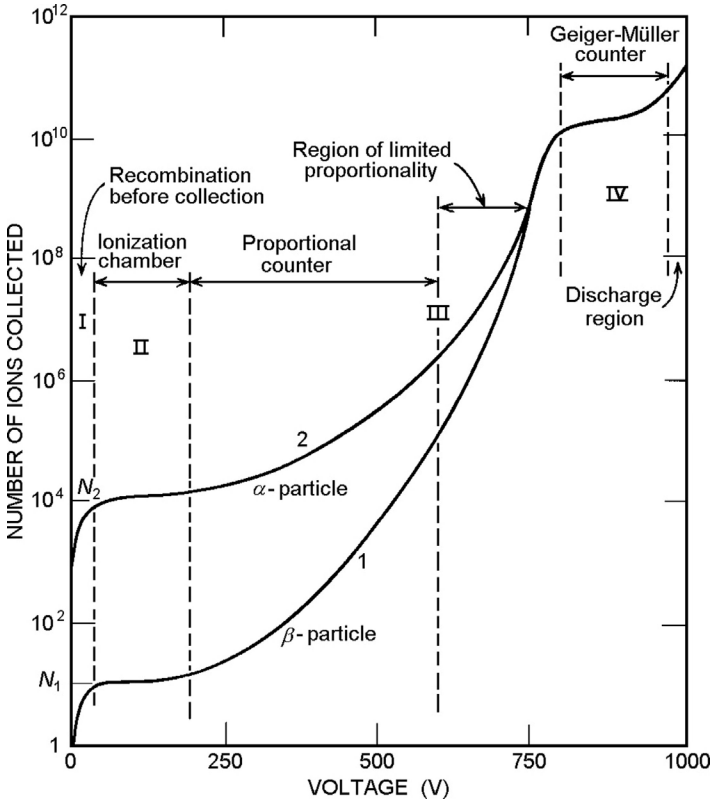


Figure 9.8 Number of ion pairs formed in a gas-filled ionization chamber with a thin wire anode as function of anode voltage. (Acc. to C. G. Montgomery and D. D. Montgomery.)

9.3.1. Ion chambers

The ion chamber is a gas-filled space between two electrodes. In Figure 9.10(a) the electrodes are two parallel plates, but another common geometrical arrangement uses the cathode as a hollow cylinder and the anode as a thin wire in its center, e.g. the GM tube in Figure 9.9(a). In other chambers the chamber walls serve as the cathode with a thin wire loop as anode as illustrated in Figure 9.9(b). The chamber may be designed for recording radiation reaching it from the outside, or it may be used for measuring radioactive samples placed within it. Some chambers have additional electrodes, usually a thin central grid to improve measuring conditions. The anode is kept at a positive potential 100—1000 V above the cathode.

Ions and electrons formed in the gas by nuclear radiation move towards the electrodes where they are discharged. If the gas is pure argon, only Ar^+ and e^- are formed. The electrons move rapidly towards the anode, and then through the electrical circuitry, as shown in Figure 9.10(b), over the resistor R towards the cathode where they neutralize

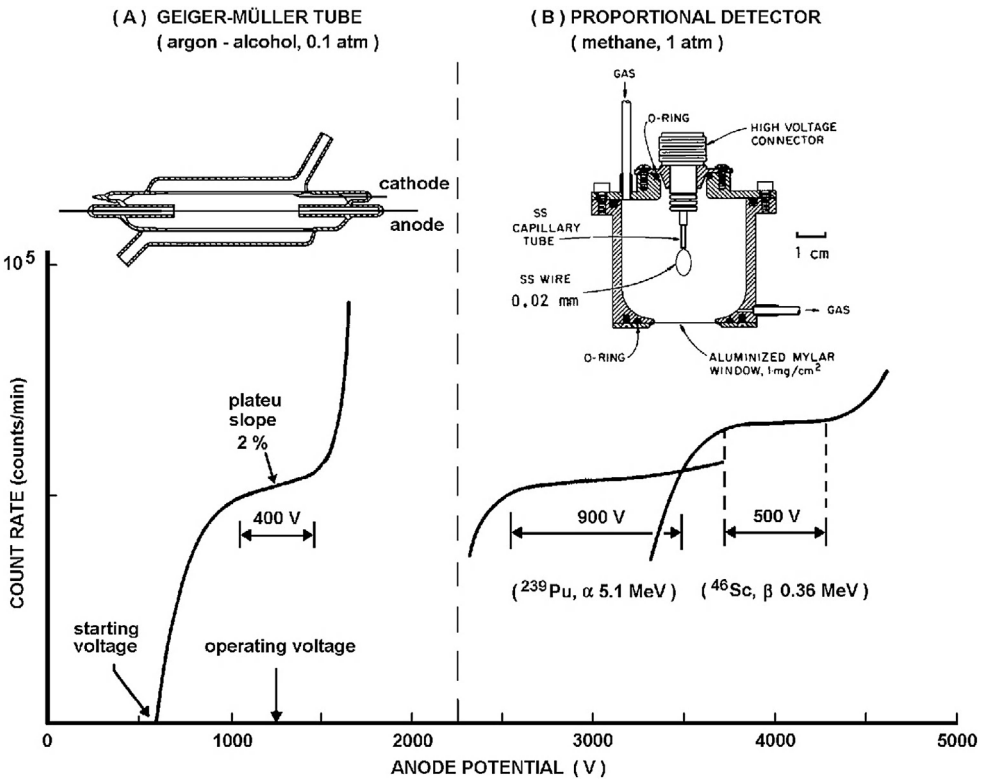


Figure 9.9 Characteristics for GM and proportional counter tubes. The GM tube (a) is designed for flowing liquids. The proportional tube (b) uses a flowing counting gas.

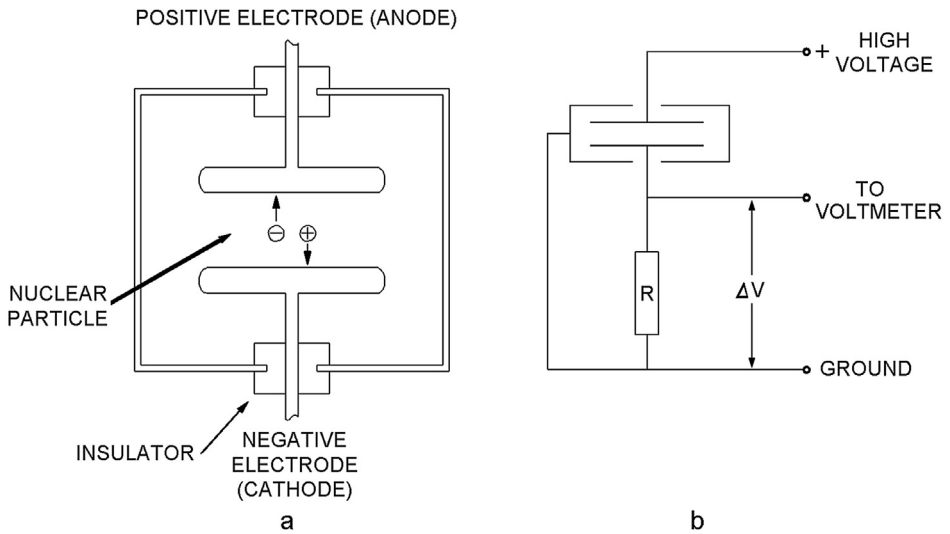


Figure 9.10 (a) A parallel plate ionization chamber and (b) its measuring circuitry.

the argon ions: $Ar^+ + e^- \rightarrow Ar$. The gas is therefore not used up. The current i through the chamber and through the resistor R causes a voltage drop V,

$$V = R\, i \tag{9.23}$$

which can be continuously recorded by a sensitive voltmeter. If the current or voltage drop, ΔV, is measured as a function of the voltage V applied over the electrodes, it is found that the current (or ΔV) increases with V up to a saturation value. The reason for this is that at low voltages some of the positive ions formed initially by the radiation recombine with the electrons, reducing the collected charge. With higher voltages the cations and electrons separate more rapidly with less recombination, and at the saturation value essentially no recombination occurs. The ionization chamber will always be operated at saturation voltage.

Suppose a 100 kBq α-sample is placed within an argon-filled chamber of sufficient size that all the 5 MeV α's emitted are stopped in the gas volume. The saturation ion current will be found to be:

$$i = 1.60 \times 10^{-19} A\, E_{\text{loss}}\, \eta\, \psi_{\text{geom}}\, w^{-1} \tag{9.24}$$

where A is the radioactivity of the sample in Bq, E_{loss} is the total energy lost per particle (5×10^6 eV) to the detector, η is the collection efficiency, ψ_{geom} is the geometric efficiency (for a thin solid sample we shall assume it to be half of a full sphere, i.e. 0.50), w is the energy required for the formation of an ion pair in the gas (for argon 26 eV), while the constant is the charge (Coulomb) of a single ion (the ion pairs must be regarded as single charges): thus $i = 1.54 \times 10^{-9}$ A. With a resistor of 10^9 Ω, the voltage drop ΔV is 1.5 V.

Two types of ion chamber are common: (i) Simple, portable, rugged instruments with resistors $\leq 10^{13}$ Ω. With these, radiation intensities of $\geq 10^3$ $\beta\ s^{-1}$ and of $\geq 10^5$ $\gamma\ s^{-1}$ can be measured. They are usually calibrated in dose rate (e.g. Gy h^{-1}) and used for radiation protection measurements under field conditions (*dose rate meters*). (ii) Advanced, very sensitive instruments, with very high resistors ($\sim 10^{15}$ Ω) or special circuitry as in the vibrating reed electrometer. The chamber must always be designed with extreme care to avoid leaking currents from the anode over the chamber casing to the cathode. One way to minimize this is to ground the casing, as shown in Figure 9.10(b). In the best of these instruments, currents as low as 10^{-18} A can be measured, corresponding to less than 1 $\alpha\ h^{-1}$, 10 $\beta\ min^{-1}$, or 10 $\gamma\ s^{-1}$. These instruments are best suited for measurement of radioactive gases, like tritium or radon in nature.

The *beta-current neutron detector* is a solid state ion chamber which is used in nuclear reactor technology. It consists of an emitter in which a nuclear reaction occurs, leading to the emission of primary β^- particles (e.g. through the reaction (^{103}Rh(n,γ)^{104}Rh(β^-, 4.2 s) ^{104}Pd) or secondary electrons (e.g. through absorption of the prompt γ's emitted in the neutron capture). These electrons represent a current and are collected by an anode. The radioactive decay type detectors have a response time depending on the product

half-life, which the capture-γ detectors lack. These detectors have a limited lifetime; for the $^{59}Co(n,\gamma)^{60}Co$ it amounts to 0.1% per month at 10^{13} n $cm^{-2}s^{-1}$. The lifetime depends on the $\sigma_{n\gamma}$ value (37 b for ^{59}Co, 146 b for ^{103}Rh).

9.3.2. Proportional counters

Values of a in Eqn (9.5) of 10^3–10^5 are commonly achieved in proportional counter operation. If $a = 10^3$, essentially all the gas multiplication occurs within 10 mean free path lengths from the wire for the electron in the gas (2^{10}=1024). At 1 atm the mean free path length is approximately 10^{-6} m, which means that the gas multiplication occurs within 0.01 mm of the wire.

The gas multiplication factor varies with the applied voltage but for a given voltage a is constant so the detector pulse output is directly proportional to the primary ionization. As a result it is possible to use a proportional counter to distinguish between α- and β-particles and between identical particles of different energies in as much as different amounts of primary ionization are produced in these cases.

The output pulse in proportional counter operation is not dependent on the collection of the positive ions by the cathode. Consequently the rate of detection depends on the time necessary for the primary electrons to drift into the region of high field strength near the anode wire. As a result, proportional counters have a much shorter resolving time than ion chambers which depend on the slow-moving positive ions. In fact the detector tube in a proportional counter can amplify a new pulse before the positive ion cloud of the previous pulse has moved very far if the new ionization occurs at a different location on the center wire. Time intervals necessary to enable the counter to measure two distinct pulses can be as low as 0.2–0.5 μs. Frequently the associated measuring equipment is a greater determinant of the resolving time than the detector itself. If, however, a proportional counter is being used for the measurement of particle energies, any residual positive ion cloud must have time to drift an appreciable distance before a new pulse is generated. In this case the resolving time is closer to 100 μs.

Counting gases consists usually of one of the noble gases mixed with a small amount of polyatomic gas. The latter makes the gas multiplication factor less dependent on applied voltage, and increases the speed of electron collection. Typical counting gas mixtures are 90% Ar + 10% CH_4 ("P-gas") and 96% He + 4% i-C_4H_{10} ("Q-gas"). Many other pure gases or combinations are possible, but molecules which readily attach electrons must be avoided. Figure 9.9(b) shows a proportional counter tube using pure CH_4 and the tube characteristics (i.e. count rate versus voltage) for ^{239}Pu α-particles and for ^{46}Sc β-particles entering through the thin aluminized mylar window.

The gas mixture and electrodes may be separated by a thin window from the radioactive sample, or the counter may be operated windowless. For windowless operation, after insertion of the sample the chamber must be flushed with gas to eliminate all oxygen and water vapour as these molecules absorb electrons readily to form negative ions and, by

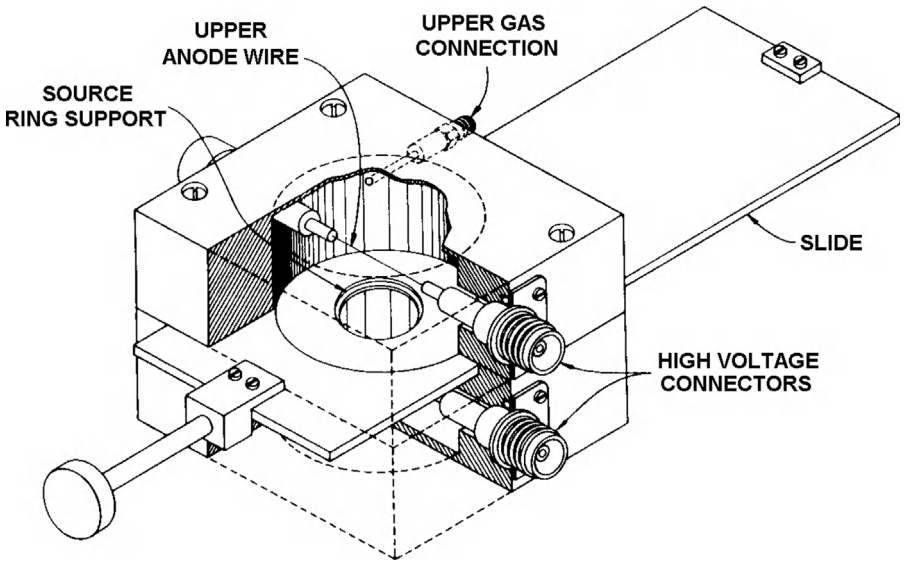


Figure 9.11 A 4π proportional counter for measuring absolute decay rates. (Acc. to O'Kelley.)

so doing reduce the pulse size. For α- and β-particles whose ranges do not exceed the dimensions of the chamber, windowless counters are often referred to as 2π counters since a solid angle of 2π is subtended above the sample. In these cases, with proper care, the measured count rate is very close to 50% of the true disintegration rate. Such windowless proportional counters are very useful for measuring low energy radiation such as the β-emissions of ^{14}C and ^{3}H, and for absolute counting. Figure 9.11 shows a 4π proportional counter in which the sample is inserted in the middle between the two half-domes (cathodes) giving near 100% geometric efficiency.

Proportional counters can be used also for *neutron detection* by using a gas containing nuclei that capture neutrons and in the nuclear reaction produce charged particles, e.g. ^{10}B or ^{3}He. In BF_3-counter the gas is BF_3, usually enriched in ^{10}B. With neutrons the reaction

$$^{10}B + n \rightarrow {}^{7}Li + {}^{4}He \quad (Q = 2.78\ \text{MeV}) \tag{9.25}$$

occurs (cf. Figure 11.4). The ionization of the two products produces a heavy pulse, which is easy to discriminate against an intense γ-background. Some properties of a BF_3 neutron detector are given in Table 9.2. From Figure 11.4 it is clear that the BF_3 counter has a higher efficiency for thermal than for fast neutrons.

Another technique for neutron detection uses a *fission chamber*. One design contains a stack of alternate anodes and cathodes, one of the electrodes being covered by a thin layer of uranium enriched in ^{235}U. The fission fragments produce large ionization even

Table 9.2 Properties of some representative counter tubes

	Geiger Müller counters		Proportional counters	
Purpose	β,γ	α,β	n	α,β
Wall thickness	0.1mm glass	1.5 mg/cm^2 mica	1 mm steel	0.3 mg/cm^2 foil
Filling gas	Ne+Ar+halogen	Ar+organic	^{10}B enriched BF_3	Pure CH_4
Operating voltage	700 V	1250 V	2200 V	α 1900 V, β 2600 V
Plateau length	>250 V	~300 V	>300 V	α > 800 V, β~400V
Plateau slope	8%/100 V	<4%/100 V	<2%/100 V	<2%/100 V
Lifetime	>3 × 10^{9}counts	3 × 10^{8}counts		Infinite
Dead time	140 s	300 s		3 s
Background count rate	20−30 cpm	80 cpm	1−2 cpm	α 0.1 cpm β 20−25 cpm
Background shielding	50 cm Pb	10 cm Pb		50 cm Pb
Special features	Insensitive to overvoltage		Insensitive to γ	

though the gas multiplication is quite low. This detector is more sensitive to fast neutrons than the BF_3 counter, and can be used for fast neutron fluxes up to ~10^{10} n m^{-2} s^{-1} with a background of a few cps.

9.3.3. Geiger-Müller counters

In region IV (Fig. 9.8), the proportionality between the primary ionization and the output pulse disappears and the latter becomes the same size for all initial ionization whether it be a 6 MeV α-particle or a 50 keV X-ray. Geiger-Müller (GM) counters which operate in this region have high sensitivity to all different kinds of radiation and the large size of the output pulse (from a tenth of a volt to one volt, compared to the several tenths of a millivolt output of ionization chambers) requires much less external amplification. This considerably reduces the complexity of the auxiliary electronic equipment. The detector tubes for GM counters are quite simple and allow a great deal of flexibility in design. Figure 9.9(a) shows a GM tube with a jacket for flowing liquids; Table 9.2 gives the properties of some other typical GM tubes. In general GM counters are limited to handling lower count rates than proportional counters.

As in the case of proportional counters, the primary electrons from the ionizing radiation cause secondary ionization near the center anode wire in GM detectors. This initial avalanche ends when the very mobile electrons are all collected by the anode. However, the neutralization of the electrons at the wire produces photons, which react

with the gas leading to the emission of photoelectrons. These trigger further avalanches and an overall avalanche spreads along the complete length of the center wire and continues until the build-up of the positive ion sheath progresses to a point sufficient to reduce the field strength sufficiently to prevent further ionization. This build-up takes place because the heavy positive ions have such a slow rate of movement that they are essentially stationary during the time interval of the electron avalanches. The time required to reach this point in the process is of the order of a few microseconds. Since the dead-time is of the order of 100 μs due to the slow movement of the heavy cations to the cathode, it is necessary to make dead-time corrections in GM counting at much lower count rates than in the case of proportional counting. Typically, corrections are significant for count rates exceeding 10 000 cpm.

When the positive ion sheath reaches near the cathode it may induce new avalanches as it collides with atoms in the gas and produces UV-light or emission of electrons. To avoid a recurring pulsing, which would render the counter useless, it is necessary to prevent further avalanches at this point by a process of quenching. This is usually accomplished by the addition of a small amount of an organic compound such as ethyl alcohol or ethyl formate or a halogen to the counting gas. Since the ionization potential of the organic molecule is lower than that of argon, the usual counting gas, when the positive argon sheath moves to the cathode and encounters organic molecules the following reaction occurs:

$$Ar^{+} + C_2H_5OH \rightarrow Ar + C_2H_5OH^{+} \tag{9.26}$$

In this reaction the charge of the argon ion is transferred to the organic molecule which gains an electron upon striking the cathode. The energy acquired in the neutralization causes dissociation into uncharged fragments rather than producing photon or electron emission. In as much as the quenching gas is dissociated in the process of counting, such GM tubes have limited lifetimes which usually amount to approximately 10^9 discharges.

Chlorine and bromine have strong absorption bands below about 251 nm for the photons emitted, leading to dissociation. In the recombination the halogen molecule returns to its ground state via a series of low energy excited states. This makes halogens useful as quenchers in GM tubes. Halogen-filled GM tubes are popular because of their infinite lifetime and low operating voltage. Another advantage is that they will not be damaged by wrong polarity or excess voltage, as is the case with the organic quenched tubes.

GM tubes are available in a wide variety of shapes and sizes. Tubes have been used successfully which varied from approximately 1 mm to several centimeters in diameter and from 1 cm to almost a meter in length. The cathode can be made by coating the inside of a glass cylinder with a conducting material such as metal or graphite while the anode may be a tungsten wire mounted coaxially. The "end-window" GM tube has a

thin mica window and the center wire is terminated with a glass bead. The cathode is always at ground potential while the anode is at a high positive potential.

When the radiation intensity is measured as a function of the electrode potential it is found for GM counters (as for proportional counters) that over a certain voltage interval there is little, if any, change in the measured count rate. This is known as the *plateau region* (Fig. 9.9). This is the voltage range in which the detectors are operated since they are relatively insensitive to small voltage changes in this region. For organic quenched GM tubes the plateau commonly occurs between 1200 and 1500 V and its slope should not exceed 5% per 100 V. Halogen-quenched tubes operate at a lower voltage but have higher plateau slopes (cf. Table 9.2).

9.4. SEMICONDUCTOR DETECTORS

The semiconductor detector is similar to an ordinary semiconductor diode composed of p-type and n-type semiconductor material. This detector has become dominant for nuclear spectroscopy (i.e. determination of the energy of nuclear radiation) but it is not so often used for simple measurement of count rates.

Semiconductors are materials like silicon (resistivity ~10^3 Ωm) and germanium (0.6 Ωm) with resistivities between metals (e.g. copper, 10^{-8} Ωm) and insulators (e.g. quartz, 10^{12} Ωm). A crystal of pure silicon placed between two electrodes is almost nonconducting. The electrons in the material are almost all *valence electrons*, bound to specific silicon atoms with an energy of 1.115 eV at room temperature (0.75 eV for Ge at 80 K). If 1.115 eV is given to an electron in silicon, it moves to a band of overlapping energy levels which are not associated with specific atoms. The electron moves readily through the crystal in this "conduction band", i.e. the crystal conducts electricity. At a certain temperature some electrons, according to the Maxwell energy distribution, always have the necessary energy to be in the conduction band. These electrons provide a very small conductivity for pure silicon; this is referred to as the *intrinsic conductivity*. For diamond, the gap between the valence and conducting band is 7 eV, which is so large that essentially no electrons are found in the conduction band at room temperature, and thus diamond is an insulator.

The energy needed for transferring valence electrons to the conduction band can be supplied by nuclear radiation. The average energy needed to produce an electron-hole pair in silicon at room temperature is not 1.115 eV but 3.62 eV because some energy is lost as crystal excitation (at 80 K the exitation energy is 3.72 eV in Si and 2.95 eV in Ge). The electron removed from the valence band leaves a vacancy or "hole". The ionization is said to give an *electron-hole pair*. Just as the electrons move towards the anode, the holes move towards the anode.

Si has 4 valence electrons while P has 5 and In 3. If we introduce a very small amount of phosphorus into silicon, the phosphorus atoms substitute for silicon in the crystal